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Introduction

The motion of solid objects in Euclidean space is an important field in geometry, physics and computer graphics. Having a mathematical model of the motion of such objects is extremely important in physics and engineering, and having a good understanding of the structure of this model is important if we wish to derive meaningful results using the model.

In this (relatively short) discussion, I will present a mathematical model for the motion of solid objects using concepts from differential calculus, group theory and linear algebra. Familiarity with those fields is assumed, but an overview of the important definitions and results is also provided whenever those are needed, so most readers with basic academic background should be able to follow along.

In the second part of the discussion, I consider a special group of rigid motions - the symmetry group - as a segue to dihedral groups, which is an important family of groups studied in abstract algebra.

Structure of $\mathbb{R}^n$

We start with a quick refresher of the inner product, norm and metric of $\mathbb{R}^n$. I have discussed these at length in Differential Calculus.

DEFINITION 0.1 $\langle$**R**^{n}$>$

$\mathbb{R}^n$ is the vector space whose vectors are the n-tuples $v = (v_1, \dots, v_n)$ where $\forall v_i \in v(v_i \in \mathbb{R})$. The standard basis of $\mathbb{R}^n$ is $e = \{(1, 0, \dots), (0, 1, \dots), (0, 0, 1, \dots), \dots, (0, \dots, 1)\}$.

We equip $\mathbb{R}^n$ with a notion of *distance*:

DEFINITION 0.2 $\langle$**metric of R**^{n}$>$

The metric $d(p, q)$ *of two points* $p = (p_1, \ldots, p_n) \in \mathbb{R}^n, q = (q_1, \ldots, q_n) \in \mathbb{R}^n$*, is given by the* Euclidean distance*:*

$$d(p, q) = |p - q| = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$$

It can be verified that the Euclidean distance is indeed a metric, i.e. it has the following properties $\forall p, q, w \in \mathbb{R}^n$:
1. $d(p, p) = 0$
2. $0d(p, q) \iff q \neq p$ (positive definiteness)
3. $d(p, q) = d(q, p)$ (symmetry)
4. $d(p, w) \leq d(p, q) + d(q, w)$ (triangle inequality)

The structure $(\mathbb{R}^n, d_{\text{euclid}})$ is then a *metric space* on $\mathbb{R}^n$. Using Euclidean distance, we define the *norm* of $\mathbb{R}^n$, thus equipping $\mathbb{R}^n$ with a notion of *length*:

DEFINITION 0.3 $\langle$**norm of R**^{n}$>$

The norm $||v||$ *of a vector* $v \in \mathbb{R}^n$ *is given by* $||v|| = d(v, 0) = |v - 0| = |v|$*, or more explicitly* $||v|| = \sqrt{\sum_{i=1}^{n} v_i^2}$*. The metric* d *is the* induced metric *of the norm, and alternatively we could have defined* d *in terms of the norm as* $d(u, v) = ||u - v||$*.*

It can be verified that the Euclidean norm is indeed a norm, i.e. it has the following properties $\forall v, u \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:
1. $||v|| = 0 \iff v = 0$, where 0 is the 0 vector in $\mathbb{R}^n$, i.e. $0 = (0, \ldots, 0)$. (positive-definite)
2. $0||v|| \iff v \neq 0$ (positive definiteness)
3. $||\lambda v|| = |\lambda| \cdot ||v||$ (absolute homogeneity)
4. $||v + u|| \leq ||v|| + ||u||$ (triangle inequality)

The structure $(\mathbb{R}^n, || \cdot ||)$ is a *normed vector space*. Consider the inner product of $\mathbb{R}^n$ which is usually taken as the dot product:

DEFINITION 0.4 $\langle$**inner product of R**^{n}$>$

The inner product $\langle u, v\rangle$ *of* $u, v \in \mathbb{R}^n$ *is given by the* dot product*, i.e.* $\langle u, v\rangle = u \cdot v = \sum_{i=1}^{n} u_i v_i$*. Alternatively, one may define the dot product via matrix multiplication by writing* $u \cdot v = u^T v$*, where* $\cdot$ *on the LHS is the dot product and the multiplication on the RHS is matrix multiplication.*

It can be shown that the dot product is indeed an inner product, i.e. it has the following properties $\forall u, v, w \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$:
1. $u \cdot v = \overline{v \cdot u}$ (conjugate symmetry) (not that since $u, v \in \mathbb{R}^n$ in our example then we have $u \cdot v = v \cdot u$ since $\overline{v \cdot u} = v \cdot u$ since the vectors don't have a complex part)
2. $(\alpha v + \beta u) \cdot w = \alpha(v \cdot w) + \beta(u \cdot w)$ (linearity in first argument)
3. $u \cdot u = 0 \iff u = 0$
4. $0 u \cdot u \iff u \neq 0$ (positive definiteness)

The structure $(\mathbb{R}^n, \cdot)$ where $\cdot$ is the dot product operator is an *inner-product space*. Every inner-product *induces* a *norm* as follows: Let V be a vector field over a scalar field $\mathbb{F}$ and $\langle , \rangle : V \times V \to \mathbb{F}$ be an inner-product operator, then $|| \cdot || : V \to \mathbb{F}, ||v|| \mapsto \sqrt{< v, v >}$ is called the *induced norm* of the inner-product (note that this operation is well defined in both $\mathbb{R}$ and $\mathbb{C}$ since the inner-product is positive definite). The reader can verify that it is indeed a norm.

EXAMPLE 0.1 $\langle$norm induced by the dot product$\rangle$

Consider $\mathbb{R}^n$ and the dot product, then given $v \in \mathbb{R}^n$ the induced norm of the dot product is given by $||v|| = \sqrt{v \cdot v} = \sqrt{\sum_{i=1}^n v_i^2}$, Observe that this is exactly the regular Euclidean norm defined in 0.3 norm of R^n.

In an inner-product space, the Pythagorean theorem holds:

THEOREM 1 $\langle$Pythagorean theorem$\rangle$

Let $(V, \langle \cdot \rangle)$ be an inner product space where $|| \cdot || : V \to \mathbb{R}$ be the induced norm, then $\forall u, v \in V$, if $\langle u, v \rangle = 0$ then $||u + v||^2 = ||u||^2 + ||v||^2$.

Proof. First note that $\langle u, v \rangle = 0 \iff \langle v, u \rangle = 0$. This follows from the conjugate symmetry: $\langle u, v \rangle = \overline{\langle v, u \rangle} = 0 \implies \langle v, u \rangle = 0$. Denote this result $(*)$. Then,

$$||u+v||^2 = \langle u+v, u+v\rangle \underset{\text{linearity}}{=}$$

$$\langle u, u+v\rangle + \langle v, u+v\rangle \underset{\text{conj. symmetry}}{=}$$

$$\overline{\langle u+v, u\rangle} + \overline{\langle u+v, v\rangle} \underset{\text{linearity}}{=}$$

$$\overline{\langle u, u\rangle} + \overline{\langle v, u\rangle} + \overline{\langle u, v\rangle} + \overline{\langle v, v\rangle} \underset{\text{by }(*)}{=}$$

$$\overline{\langle u, u\rangle} + \overline{\langle v, v\rangle} =$$

$$||u||^2 + ||v||^2$$

□

DEFINITION 0.5 $\langle$orthogonal vectors$\rangle$

If $\langle u, v\rangle = 0$ in an inner-product space, we say that u and v are orthogonal*. (note that the order of u, v is not important since by the previous proof $\langle u, v\rangle = 0 \iff \langle v, u\rangle = 0$)*

PROPOSITION 1

Let $u, v \in V$, V be an inner-product vector space with induced norm, and $v \neq 0$, then $z = u - \frac{\langle u,v\rangle}{||v||^2}v$ is orthogonal to v. Furthermore, it is orthogonal to all vectors of the form αv where $\alpha \in \mathbb{F}, \alpha \neq 0$.

Proof. Note that z is defined since $v \neq 0 \implies ||v|| \neq 0$. We prove by direct calculation of $\langle z, v\rangle$.

$$\langle z, v\rangle = \langle u - \frac{\langle u, v\rangle}{||v||^2}v, v\rangle = \langle u, v\rangle - \frac{\langle u, v\rangle}{||v||^2}\langle v, v\rangle = \langle u, v\rangle - \langle u, v\rangle = 0$$

For the second part of the proof, recall $\langle z, \alpha v\rangle = 0 \iff \langle \alpha v, z\rangle = 0$, which by linearity expands to $\alpha\langle v, z\rangle = \alpha \cdot 0 = 0$.

□

In any inner-product space, the Cauchy-Schwarz inequality always holds:

THEOREM 2 $\langle$Cauchy-Schwarz (CS) inequality$\rangle$

Let $(V, \langle\cdot\rangle)$ be an inner-product space and $||\cdot||$ be the induced norm, then $\forall u, v \in V$:

$$|\langle u, v\rangle| \leq ||u|| \cdot ||v||$$

Moreover, if $v \neq 0$ then $\langle u, v\rangle = ||u|| \cdot ||v||$ if and only if u, v are linearly dependent.

Proof. We start with the trivial case where $v = 0$, then we have $\langle u, v\rangle = \langle u, 0\rangle = \overline{\langle 0, u\rangle}$. Write $0 = 0u$, then $\overline{\langle 0, u\rangle} = \overline{0\langle u, u\rangle} = 0$. Similarly, $v = 0 \iff ||v|| = 0$, so $||u|| \cdot ||v|| = 0$, and we get $0 = |\langle u, v\rangle| = ||u|| \cdot ||v|| = 0$, so the inequality holds.

Suppose then that $v \neq 0$, then take $z = u - \frac{\langle u,v\rangle}{||v||^2}v = u - \alpha v$ for some $\alpha \in \mathbb{R}, 0\alpha$, then by 1 $\langle z, v\rangle = 0$ and also $\langle z, \alpha v\rangle = 0$, so by Theorem 1 Pythagorean theorem we have $||z + \alpha v||^2 = ||z||^2 + ||\alpha v||^2$, but $z = u - \alpha v$ so $z + \alpha v = u$, so $||u||^2 = ||z||^2 + ||\alpha v||^2$, so:

$$||u||^2 = ||z||^2 + ||\alpha v||^2 = \alpha^2||v||^2 + ||z||^2 = \frac{\langle u, v\rangle^2}{||v||^4}||v||^2 + ||z||^2 = \frac{\langle u, v\rangle^2}{||v||^2} + ||z||^2$$

Multiplying both sides by $||v||^2$, we get

$$||u||^2 \cdot ||v||^2 = \langle u, v\rangle^2 + ||z||^2 \cdot ||v||^2$$

Since $0 \leq ||z||$ by definition and $0||v||$ by supposition $v \neq 0$, it follows that $(||u|| \cdot ||v||)^2 = ||u||^2 \cdot ||v||^2 \geq \langle u, v\rangle^2$. Taking the square-root of the inequality, we retrieve the Cauchy-Schwarz (CS) inequality $|\langle u, v\rangle| \leq ||u|| \cdot ||v||$.

Furthermore, clearly the inequality becomes an equality supposing $v \neq 0$ if and only if $||z|| = 0$, which is only true when $z = 0$, i.e. $u = \frac{\langle u,v\rangle}{||v||^2}v$, i.e. u and v are linearly dependent. It remains to show the converse, i.e. that $u = \alpha v$ for some $\alpha \in \mathbb{F}$ implies $|\langle u, v\rangle| = ||u|| \cdot ||v||$, and indeed in this case we get $\langle u, v\rangle = \alpha\langle v, v\rangle = \alpha||v||^2 = \alpha||v|| \cdot ||v||$, and taking the absolute value we get $|langleu, v\rangle = ||\alpha v|| \cdot ||v|| = ||u|| \cdot ||v||$, so $|\langle u, v\rangle| = ||u|| \cdot ||v||$ if and only if (u, v) are linearly dependent.

$\square$

COROLLARY 2.1

Suppose $v \neq 0$, then:

$$0 \leq \frac{|\langle u, v\rangle|}{||u|| \cdot ||v||} \leq 1$$

Proof. Immediately from the theorem by dividing both sides by $||u|| \cdot ||v||$. □

By 2.1 we can use the inner-product to measure *angles* as follows:

DEFINITION 0.6 $\langle$angle between two vectors in an inner-product space$\rangle$

Let u, v be two vectors in an inner-product space with the induced norm, and let $\theta \in [0, \pi]$ be the angle between u and v, then θ is the solution to the following equation in the range $[0, \pi]$:

$$\cos\theta = \frac{|\langle u, v\rangle|}{||u|| \cdot ||v||}$$

Alternatively, we write $|\langle u, v\rangle| = ||u|| \cdot ||v|| \cdot \cos\theta$.

This is in line with the geometric interpretation of the dot product in $\mathbb{R}^2, \mathbb{R}^3$, and can be taken as the definition in $\mathbb{R}^n$ for the case $3n$. One can also show that $u \cdot v = ||u|| \cdot ||v|| \cdot \cos\alpha$ in $\mathbb{R}^2, \mathbb{R}^3$ using analytic geometry using the geometrical definition for an angle, thus the two definitions are in fact equal. We will use this equivalence in the next section.

Isometries of $\mathbb{R}^n$

Recall that an *isometry* is a distance preserving map, and that all isometries are injective (this is trivial to prove by contradiction). Given a metric space (V, d), one can demonstrate that the isometries of V to itself are a group under composition:

PROPOSITION 2 $\langle$Isometry group of a metric space$\rangle$

Let (V, d) be a metric space, then the isometries $\phi : V \to V$ form a group under composition. We denote this group Isom(V).

Proof. We will show that the structure $(\Omega, \circ)$ where Ω is the set of isometries of V and $\circ$ is function composition satisfies all group axioms.

1. (closure) Let ϕ, τ be isometries and consider $\psi = \phi \circ \tau$. Since both are maps on V, the composition is

well-defined. Now let $u, v \in V$ and consider $d(\psi(u), \psi(v))$. Since $\psi(u) = \phi(\tau(u))$ and $\psi(v) = \phi(\tau(v))$, then we have $d(\psi(u), \psi(v)) = d(\phi(\tau(u)), \phi(\tau(v)))$, now since ϕ is an isometry then by definition we have that the last expression is equal to $d(\tau(u), \tau(v))$, and since τ is also an isometry then this expression is equal to $d(u, v)$, so $d(\psi(u), \psi(v)) = d(u, v)$, so ψ is an isometry, so the set of isometries is closed under composition.
2. (identity) Identify that the identity map $\mathrm{id}_V : V \to V, v \mapsto v$ is an isometry since $d(\mathrm{id}_V(v), \mathrm{id}_V(u)) = d(v, u)$, so id is in the set of isometries of V. Next, identify that $\phi \circ \mathrm{id}_V = \mathrm{id}_V \circ \phi = \phi$, so the identity map is the identity element of the set of isometries under composition.
3. (left-Inverse) Let $\phi \in \Omega$, then ϕ is an isometry and thus injective, so it has a left-inverse ψ s.t. $(\psi \circ \phi)(x) = x$, i.e. $\psi \circ \phi = \mathrm{id}_V$, which is the identity element on $(\Omega, \circ)$. (every injection $f : A \to B$ has a left-inverse $g : B \to A$ which can be constructed explicitly as follows: $g = \{(y, x) | y \in B \land x \in A \land f(x) = y\}$, clearly g is a relation from B to A, so to show it is well-defined we suppose by contradiction it is not, i.e. $\exists y \in B$ s.t. $\exists x_1, x_2 \in A, x_1 \neq x_2$ and $(y, x_1), (y, x_2) \in g$, but then $f(x_1) = y = f(x_2)$ and since f is injective it must be that $x_1 = x_2$, a contradiction, so g is well-defined)
4. (associativity) mapping (function) composition is associative so it is also associative on the set of isometries.

We conclude that the structure $(\mathrm{Isom}(V), \circ)$ is a group.

$\square$

Consider an inner-product space $(V, \langle\cdot\rangle)$ with induced norm and metric, then the following is true:

THEOREM 3

Let $(V, \langle\cdot\rangle)$ be an inner-product space s.t. $\langle u, v\rangle = \langle v, u\rangle$ with induced norm and metric and $\phi : V \to V$ be an isometry, then ϕ preserves the inner product if and only if $\phi(0) = 0$, i.e. ϕ fixes the origin.

Proof. $\implies$ First observe that $d(\phi(0), 0) = |\phi(0) - 0| = ||\phi(0)|| = \sqrt{\langle\phi(0), \phi(0)\rangle}$ by definition. Now, since ϕ preserves the inner product, we have $\langle\phi(0), \phi(0)\rangle = \langle 0, 0\rangle$, which is 0 by positive definiteness of the inner product, so $\sqrt{\langle\phi(0), \phi(0)\rangle} = \sqrt{0} = 0$, so again by positive definiteness we have $\phi(0) = 0$. (note that for this part of the proof we did not use the fact that ϕ is an isometry)

$\impliedby$ Suppose ϕ is an isometry that fixes the origin and let $u, v \in V$. Since ϕ is an isometry then $\forall w \in V, d(\phi(w), \phi(0)) = d(w, 0) = ||w|| = \sqrt{\langle w, 0\rangle}$, and since ϕ fixes the origin then $d(\phi(w), \phi(0)) = d(\phi(w), 0) = d(\phi(w), 0) = ||\phi(w)|| = \sqrt{\langle\phi(w), 0\rangle}$, so $||w|| = ||\phi(w)||$, and in particular $||\phi(v)|| = ||v||$ and $||\phi(u)|| = ||u||$.

Consider the vector $\phi(v) - \phi(u)$ and expand its inner product with itself via linearity:

$$\begin{gathered}(*) = \\ \langle \phi(v) - \phi(u), \phi(v) - \phi(u) \rangle = \\ \langle \phi(v), \phi(v) - \phi(u) \rangle - \langle \phi(u), \phi(v) - \phi(u) \rangle = \\ \langle \phi(v) - \phi(u), \phi(v) \rangle - \langle \phi(v) - \phi(u), \phi(u) \rangle = \\ \langle \phi(v), \phi(v) \rangle - \langle (\phi(u), \phi(v) \rangle - \langle \phi(v), \phi(u) \rangle + \langle \phi(u), \phi(u) \rangle = \\ ||\phi(v)||^2 - 2\langle \phi(u), \phi(v) \rangle + ||\phi(u)||^2 = \\ (**)\end{gathered}$$

Now on the one hand, we have $(**) = ||v||^2 + ||u||^2 - 2\langle \phi(u), \phi(v) \rangle$, but on the other hand we have $(*) = d(\phi(v), \phi(u))^2$ which since ϕ is an isometry is the same as $d(v, u)$, and exapending $\langle v - u, v - u \rangle$ we retrieve a similar expression: $\langle v - u, v - u \rangle = ||v||^2 + ||u||^2 - 2\langle u, v \rangle$, which is equal to $(**)$, so $||v||^2 + ||u||^2 - 2\langle u, v \rangle = ||v||^2 + ||u||^2 - 2\langle \phi(u), \phi(v) \rangle$, so $\langle u, v \rangle = \langle \phi(u), \phi(v) \rangle$, so the inner-product is preserved.

□

In fact, by Theorem 3 we can classify *all* isometries of V with symmetric inner-product:

COROLLARY 3.1

Let $(V, \langle \cdot \rangle)$ be an inner product space and let the inner-product operator be symmetric. All isometries $\phi : V \to V$, $\exists \psi : V \to V$ s.t. ψ is an isometry that fixes the origin and $\forall v \in V(\phi(v) = \psi(v) + \phi(0)$.

Proof. Suppose ϕ is an isometry and define $\psi : V \to V$ as $\psi(v) = \phi(v) - \phi(0)$. Let $u, v \in V$. Since ϕ is an isometry, then $d(\phi(u), \phi(v)) = |\phi(u) - \phi(v)| = |u - v|$ (where $| \cdot |$ is the induced metric of the induced norm of the inner product). Observe that $d(\phi(u) - \phi(0), \phi(v) - \phi(0)) = |\phi(u) - \phi(0) - (\phi(v) - \phi(0)) = |\phi(u) - \phi(v)| = |u - v|$, but this is exactly $d(\psi(u), \psi(v))$, so by definition ψ is an isometry, and trivially it fixes the origin since $\psi(0) = \phi(0) - \phi(0) = 0$. Rearranging $\psi(v) = \phi(v) - \phi(0)$ we find $\phi(v) = \psi(v) + \phi(0)$, where ψ is an origin-fixing isometry. □

Now consider the dot product on $\mathbb{R}^n$. By 0.6 angle between two vectors in an inner-product space it follows that the angle between $u, v \in \mathbb{R}^n$ is defined by $u \cdot v$, and by Theorem 3 it follows that if ϕ is an isometry on $\mathbb{R}^n$ then it preserves the dot product if and only if it fixes the origin, so then it preserves *angles* if and only if it fixes the origin. Such isometries correspond with our intuitive notation of *rotation* of a rigid body (*rigid* in the sense that it maintains its shape, i.e. angles, lengths and distances are preserved). These isometries have a nice property - they are all *linear bijections*:

LEMMA 3.1

Let $\langle , \rangle$ be an inner-product on a vector space V over a field of real scalars, then $\forall u, v, w \in V$:

$$\langle u+v+w, u+v+w \rangle = ||u||^2 + ||v||^2 + ||w||^2 + 2(\langle u,v \rangle + \langle u,w \rangle + \langle v,w \rangle)$$

i.e. the inner product of the sum of three vectors with itself is given by the sum of the squared norms of each vector and twice the sum of the inner products of each unique combination of the three vectors.

Proof. We prove by direct calculation using the properties of the inner-product: let $u, v, w \in V$ and suppose $\overline{u} = u$, then:

$$\langle u+v+w, u+v+w \rangle \underset{\text{linearity}}{=}$$

$$\langle u, u+v+w \rangle + \langle v, u+v+w \rangle + \langle w, u+v+w \rangle \underset{\text{conj. symmetry}}{=}$$

$$\langle u+v+w, u \rangle + \langle u+v+w, v \rangle + \langle u+v+w, w \rangle \underset{\text{linearity}}{=}$$

$$\langle u,u \rangle + \langle v+w, u \rangle + \langle v,v \rangle + \langle u+w, v \rangle + \langle w,w \rangle + \langle u+v, w \rangle \underset{\text{induced norm, linearity}}{=}$$

$$||u||^2 + ||v||^2 + ||w||^2 + \langle v,u \rangle + \langle w,u \rangle + \langle u,v \rangle + \langle w,v \rangle + \langle u,w \rangle + \langle v,w \rangle \underset{\text{conj. symmetry}}{=}$$

$$||u||^2 + ||v||^2 + ||w||^2 + 2(\langle v,u \rangle + \langle u,w \rangle + \langle v,w \rangle)$$

□

THEOREM 4

Let ϕ be an origin-fixing isometry on $\mathbb{R}^n$, then ϕ is a non-singular linear transformation.

Proof. We break the proof to two lemmas:
1. ϕ is linear
2. ϕ is a bijection

From linear algebra, we know that a linear transformation is non-singular if and only if it is bijective, so by 1 and 2 we can deduce the proposition we wish to prove.

1. Let $u, v \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$, then we need to show $\phi(\lambda u + v) = \lambda\phi(u) + \phi(v)$. Consider $w = \phi(\lambda u + v) - \lambda\phi(u) - \phi(v)$, we wish to show w is zero. We do this by demonstrating that $\langle w, w\rangle = 0$, then by the properties of the inner product it will follow that $w = 0$, then it will follow that $\phi(\lambda u+v) = \lambda\phi(u)+\phi(v)$. We will use 3.1 to demonstrate this with $u' = \phi(\lambda u + v), v' = -\lambda\phi(u)$ and $w = -\phi(v)$.

First, we calculate $\langle u', v'\rangle = \langle v', u'\rangle = \langle \phi(\lambda u + v), -\lambda\phi(u)\rangle$. Recall that by Theorem 3 , since ϕ fixes the origin then it preserves the inner-product, then:

$$\begin{gathered}
\langle \phi(\lambda u + v), -\lambda\phi(u)\rangle \underset{\text{conj. symmetry, linearity}}{=} \\
-\lambda\langle \phi(u), \phi(\lambda u + v)\rangle \underset{\phi \text{ preserves } \langle\cdot\rangle}{=} \\
-\lambda\langle u, \lambda u + v\rangle \underset{\text{conj. symmetry, linearity}}{=} \\
-\lambda \cdot \lambda\langle u, u\rangle - \lambda\langle v, u\rangle = -\lambda^2||u||^2 - \lambda\langle v, u\rangle
\end{gathered}$$

Next, we calculate $\langle u', w\rangle = \langle \phi(\lambda u + v), -\phi(v)\rangle$. We get:

$$\langle \phi(\lambda u + v), -\phi(v)\rangle = -\langle \phi(\lambda u + v), \phi(v)\rangle = -\langle \lambda u + v, v\rangle = -\lambda\langle u, v\rangle - ||v||^2$$

Finally, for $\langle v', w\rangle = \langle -\lambda\phi(u), -\phi(v)\rangle$ we have

$$\langle -\lambda\phi(u), -\phi(v)\rangle = \lambda\langle \phi(u), \phi(v)\rangle = \lambda\langle u, v\rangle$$

Next, we calculate $\langle u', v'\rangle + \langle u', w\rangle + \langle v', w\rangle$:

$$\begin{gathered}
\langle u', v'\rangle + \langle u', w\rangle + \langle v', w\rangle = \\
-\lambda^2||u||^2 - \lambda\langle v, u\rangle - \lambda\langle u, v\rangle - ||v||^2 + \lambda\langle u, v\rangle = \\
-\lambda^2||u||^2 - ||v||^2 - \lambda\langle u, v\rangle = (*)
\end{gathered}$$

Next, we calculate $||u'||^2 + ||v'||^2 + ||w||^2$, recalling that since ϕ is an isometry then it preserves the norm which induces the metric, so

$$\begin{gathered}
||u'||^2 + ||v'||^2 + ||w||^2 = \\
||\phi(\lambda u + v)||^2 + || - \lambda\phi(u)||^2 + || - \phi(v)||^2 \underset{\text{norm preserved by } \phi\text{, absolute homogeneity}}{=} \\
||\lambda u + v||^2 + \lambda^2||u||^2 + ||v||^2
\end{gathered}$$

Expending $||\lambda u+v||^2=\langle \lambda u+v,\lambda u+v\rangle$ by 3.1 , we get

$$\langle \lambda u+v,\lambda u+v\rangle=||\lambda u||^2+||v||^2+2\langle \lambda u,v\rangle=\lambda^2||u||^2+||v||^2+2\lambda\langle u,v\rangle$$

So

$$||u'||^2+||v'||^2+||w||^2=2(\lambda^2||u||^2+||v||^2+\lambda\langle u,v\rangle)=(**)$$

By 3.1 and $(*),(**)$, we have

$$\langle \phi(\lambda u+v)-\lambda\phi(u)-\phi(v),\langle \phi(\lambda u+v)-\lambda\phi(u)-\phi(v)\rangle=(**)+2\cdot(*)=$$
$$2(\lambda^2||u||^2+||v||^2+\lambda\langle u,v\rangle)+2(-\lambda^2||u||^2-||v||^2-\lambda\langle u,v\rangle)=0$$

So by the properties of the inner-product, $\phi(\lambda u+v)-\lambda\phi(u)-\phi(v)=0$, so $\phi(\lambda u+v)=\lambda\phi(u)+\phi(v)$, so ϕ is a *linear*.

2. All isometries are injective (as stated earlier, this can be shown via contradiction - suppose ϕ is not injective, then $\phi(v)=\phi(u)$ for some $v\neq u$, so $d(\phi(v),\phi(u))=0$ and also $d(v,u)\neq 0$ since they are not equal, but since ϕ is an isometry then $d(\phi(v),\phi(u))=d(v,u)$, a contradiction, so ϕ is injective). By the Rank-nullity theorem, we have that for every linear transformation $T:V\to W$ where V,W are finite-dimensional vector spaces, then if T is injective *or* surjective, then it is bijective. Since $\mathbb{R}^n$ has finite dimension n, and since ϕ is a linear transformation by (1) and also injective (as an isometry), then ϕ is bijective.

□

By the previous theorem, it follows that each angle-preserving isometry on $\mathbb{R}^n$ can be represented by its transformation matrix $T\in\mathbb{R}^{n\times n}$. We wish to characterize these matrices.

Recall that a square matrix A is *orthogonal* if its transpose is its inverse, i.e. $A^T=A^{-1}$, or equivalently if $A^TA=I$.

THEOREM 5

A linear transformation preserves the dot product if and only if its transformation matrix is orthogonal.

Proof. $\implies$ Suppose A is a linear transformation on $\mathbb{R}^n$ preserving the dot product, then $u^Tv=u\cdot v=(Au)\cdot(Av)=(Au)^T(Av)$. By the properties of the transpose and by the fact that matrix multiplication is associative, we have $u^T v = (Au)^T(Av) = u^T A^T Av = u^T(A^T A)v$, so it follows that $A^T A = I$, so $A^T = A^{-1}$, so A is orthogonal.

$\Longleftarrow$ Suppose A is an orthogonal matrix, then $(Au) \cdot (Av) = (Au)^T(Av) = u^T(A^T A)v$, and since A is orthogonal then $A^T A = I$ so $u^T(A^T A)v = u^T v = u \cdot v$, so A preserves the dot product.

□

By Theorem 5 it follows that the subset of Isom$(\mathbb{R}^n)$ that preserves angles is represented by orthogonal matrices. This motivates discussing about the orthogonal subset of Isom$(\mathbb{R}^n)$.

DEFINITION 0.1 $\langle$orthogonal isometries$\rangle$

Orth$(n, \mathbb{R})$ is the set of isometries on $\mathbb{R}^n$ that preserve the dot product, i.e. (by Theorem 4) Orth$(n, \mathbb{R}) = \{\phi \in$ Isom$(\mathbb{R}^n) | \phi(0_{\mathbb{R}^n} = 0\}$.

PROPOSITION 3

Orth$(n, \mathbb{R})$ is a subgroup of Isom$(\mathbb{R}^n)$:

$$\text{Orth}(n, \mathbb{R}) \leqslant \text{Isom}(\mathbb{R}^n)$$

Proof. Clearly Orth$(n, \mathbb{R})$ is a subset of Isom$(\mathbb{R}^n)$, so we can prove the theorem by the one-step subgroup test: we show that Orth$(n, \mathbb{R})$ is non-empty, and then we show that $\forall \phi, \psi \in$ Orth$(n, \mathbb{R})$, $\phi \circ \psi^{-1} \in$ Orth$(n, \mathbb{R})$.

We show Orth$(n, \mathbb{R})$ is nonempty by showing that id$_{\mathbb{R}^n} \in$ Orth$(n, \mathbb{R})$. Recall that the identity map corresponds to the identity matrix I, and $I^T I = I$, so by Theorem 5 it is orthogonal, and we have seen that id$_{\mathbb{R}^n}$ is an isometry in the proof of 2 Isometry group of a metric space, so id$_{\mathbb{R}^n} \in$ Orth$(n, \mathbb{R})$, so it is a nonempty subset of Isom$(\mathbb{R}^n)$.

Now, suppose $\phi, \psi \in$ Orth$(n, \mathbb{R})$. Denote their corresponding transformation matrices $[\phi], [\psi] \in \mathbb{R}^n$. Since $[\psi]$ is an orthogonal matrix then it is non-singular and $[\psi^{-1}] = [\psi]^T$. Now, consider $\tau = \phi \circ \psi^{-1}$, which is also a linear transformation by Theorem 4 so its corresponding matrix is $[\tau] = [\phi][\psi^{-1}] = [\phi][\psi]^T$. Observe that

$$[\tau][\tau]^T = ([\phi][\psi]^T)([\phi][\psi]^T)^T = [\phi][\psi]^T[\psi][\phi]^T = [\phi]([\psi]^T[\psi])[\phi]^T = [\phi]I[\phi]^T = [\phi][\phi]^T = I$$

So $[\tau][\tau]^T = I$, so $[\tau]$ is orthogonal, so $\tau \in$ Orth$(n, \mathbb{R})$.

Thus by the one-step subgroup test we conclude Orth$(n, \mathbb{R}) \leqslant$ Isom$(\mathbb{R}^n)$.

□

By considering the matrices corresponding to elements of the orthogonal isometries group directly, we can also find a subgroup of the non-singular $n \times n$ matrices:

PROPOSITION 4 ⟨orthogonal group⟩

O$(n, \mathbb{R}) = \{[\phi] | \phi \in$ Orth$(n, \mathbb{R})\}$ is a subgroup of GL$(n, \mathbb{R})$, i.e. the group of non-singular matrices $A \in \mathbb{R}^{n \times n}$.

Proof. Analogous to 3 . □

Next, recall that by 3.1 , every isometry ϕ of $\mathbb{R}^n$ can be written as $\phi(v) = \psi(v) + \phi(0)$ where $\psi(v)$ is origin-fixing, and as we have seen such ψ is an orthogonal isometry representable via an orthogonal matrix A, so we have $\phi(v) = Av + v_0$ where $v_0 = \phi(0)$, so:

LEMMA 5.1

All isometries of $\mathbb{R}^n$ are affine maps.

Using this fact, we can classify the isometries of $\mathbb{R}^n$:

THEOREM 6 ⟨classification of isometries of R^n>

Let $\phi : \mathbb{R}^n \to \mathbb{R}^n$, then ϕ is either:
*1. Linear with corresponding matrix $[\phi]$ having determinant 1. (*Rotation*)*
*2. Linear with corresponding matirx $[\phi]$ having determinant -1. (*Reflection*)*
*3. $\phi : v \mapsto v + u$ for some $u \in \mathbb{R}^n$. (*Translation*)*
*4. A combination of either 1 or 2, and 3, i.e. a composition of a translation and a rotation/reflection. (*Glide reflection*)*

Proof. By 5.1 , we know ϕ is an affine map, i.e. $\exists A \in \mathbb{R}^{n \times n}, u \in \mathbb{R}^n$ s.t. $\phi(v) = Av + u$. Consider four cases:
1. $u = 0$, then ϕ is linear and by 4 orthogonal group we know $A \in O(n, \mathbb{R})$ so $A^T A = I$, so $\det A^T A = \det I = 1$, and $\det A^T A = \det A^T \det A = (\det A)^2$, so $\det A = \pm 1$ (.this is true for all orthogonal matrices). If $\det A = 1$, we say that ϕ is a *rotation* (geometrically a positive determinant preserves orientation, so it corresponds to our geometrical idea of a rotation)
2. If $u = 0$ and $\det A = -1$, that we say ϕ is a *reflection* (a negative determinant inverts orientation, which corresponds to our geometrical idea of a reflection)

3. If $A = I$, then $\phi(v) = Iv + u = v + u$, so every vector is *translated* by u under ϕ, and we call such map a *translation*. (If $u = 0$ as well, then ϕ is just the identity translation)
4. $A \neq I$ and $u \neq 0$, then consider the isomteries $\tau : v \mapsto Av$ and $\psi : v \mapsto u + v$, then $(\psi \circ \tau)(v) = \psi(\tau(v)) = \psi(Av) = Av + u = \phi$, and we call it a *glide reflection/rotation*. Note that composition here does *not* commute: evidently $\tau \circ \psi = Av + Au$, which in general $\neq Av + u = \psi \circ \tau$.

□

Using this classification, we can define Isom(R^n) as follows:

DEFINITION 0.2 ⟨Euclidean group⟩

All isometries on $\mathbb{R}^n$ given the usual Euclidean dot-product form the Euclidean group $E(n)$. *$E(n)$ consists of all translations, reflections and rotations of $\mathbb{R}^n$ and the group operation is composition.*

In fact, we can also show that *translations* form a group under composition:

PROPOSITION 5

Translations from a group under composition.

Proof. Since translations are a subset of $E(n)$, we can prove by the one-step subgroup test. First, observe that the identity isometry is a translation, so it is a non-empty subset. Now, consider two translations $t_1 : v \mapsto v + u_1, t_2 : v \mapsto v + u_2$. It is trivial to see that t_2^{-1} is given by $v \mapsto v - u_2$ (since their composition is $v \mapsto v$ which is the identity element). Now, consider $h = t_1 \circ t_2^{-1}$, we have $h(v) = t_1(t_2^{-1}(v)) = t_1(v - u_2) = (v - u_2) + u_1 = v + (u_1 - u_2)$, which is also a translation by our definition of a translation, so by the one-step subgroup test, translations form a group under composition, that is also a subgroup of $E(n)$. □

Rigid Motions as Matrices

In physics, a *rigid body* is a solid object that isn't deformed. One way to model a rigid body is via a set of point masses that comprise the body, along with the constraint that the distance between any two point masses $m_1, m_2 \in M$, where M is the rigid body (i.e. a set of point masses), remains fixed. *Rigid motion* is then the term used to describe the *motion* of a *rigid body*. Rigid motion is constrained such that it must preserve distances in the body, so it must correspond to an *isometry*.

By Theorem 6 classification of isometries of R^n. we know that all isometries can be written as the composition of a translation and an orthogonal isometry, i.e. if ϕ is an isometry, then $\phi(v) = Av + u$ where $A \in$ O$(n, \mathbb{R})$ and $v \in \mathbb{R}^n$. Recall that all affine maps of an n-dimensional vector space can be expressed as a linear map of an $n+1$-dimensional vector space over the same field of scalars via a lifting:

$$\begin{pmatrix} A & u \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v \\ 1 \end{pmatrix} = \begin{pmatrix} Av + u \\ 1 \end{pmatrix} = \begin{pmatrix} \phi(v) \\ 1 \end{pmatrix}$$

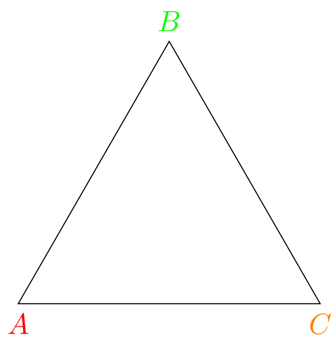
Thus we can express every rigid motion via a matrix. Also, clearly $\phi^{-1}(v) = A^T v - u$ (since ϕ is the composition of $(v \mapsto Av) \circ (v \mapsto v + u)$, and in general $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ if all inverses exist), so given a rigid motion $\begin{pmatrix} A & u \\ 0 & 1 \end{pmatrix}$, its inverse is $\begin{pmatrix} A^T & -u \\ 0 & 1 \end{pmatrix}$.

As for A, by Theorem 5 it can be any orthogonal matrix, i.e. every matrix s.t. $|\det A| = 1$. The constraint that these matrices be orthogonal reduces the degrees of freedom (DOFs) of these matrices. For example, in $O(3, \mathbb{R})$, there are only 3 DOFs, and in $O(2, \mathbb{R})$ we only have 1 DOF.

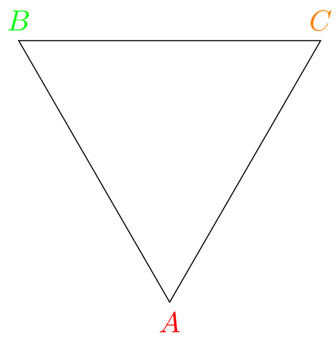
This is a very useful result, since linear transformations are composed by multiplying the corresponding transformation matrices, so the fact that rigid motions are representable via matrices that are also easily invertible is extremly useful when dealing with such motions.

Symmetry

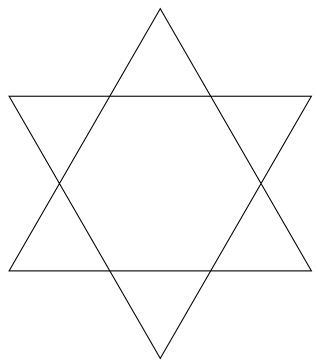
Consider an equilateral triangle $\triangle ABC$ in the Euclidean plane:



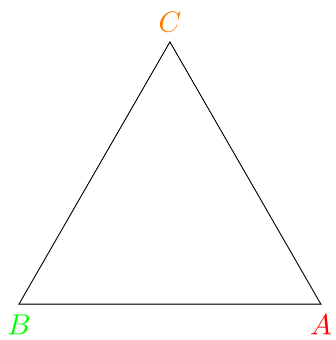
Consider $\triangle ABC$ under different plane isometries. Rotating the triangle by 60 CCW around its center, we get



Stacking both drawing on top of one another, we get



It is evident that the triangles do not overlap. However, if we rotate the triangle by 120 around its center, we get



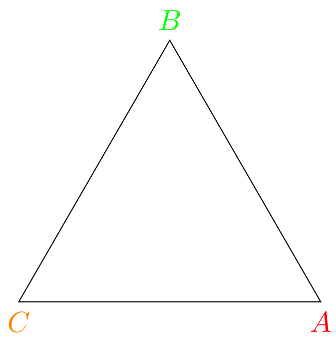
Which evidently overlaps with the original triangle. We call transformations that preserve geometrical shapes *symmetries*.

DEFINITION 0.1 ⟨symmetry⟩

Consider a geometric shape Π in a metric space V and a mapping $\phi : V \to V$. If the shape is invariant under the mapping, i.e. Im$_\phi(\Pi) = \Pi$ and ϕ preserves the structure Π, then we say that Φ is a symmetry *of Π. More generally, a symmetry of a structure A is an automorphism of A, i.e. a structure-preserving bijection from A to itself.*

It is trivial to show that the symmetries of an object form a group under composition:
1. (identity) The identity map is a symmetry, and it is also the identity under composition.
2. (associativity) Composition is associative
3. (closure) If ϕ, ψ are automorphisms of P_n, then $\tau(P_n) = (\phi \circ \psi)(P_n) = \phi(\psi(P_n)) = \phi(P_n) = P_n$ and trivially the structure of P_n is preserved under composition of automorphisms.
4. (inverse) Since ϕ is injective it has a left-inverse.

We identify 5 main types of symmetry:
1. *Rotational (radial) symmetry* is a symmetry where ϕ is a rotation. The example above displayed a rotational symmetry of an equilateral triangle.
2. *Reflection (line) symmetry* is a symmetry where ϕ is a reflection. For example, reflecting $\triangle ABC$ around an its altitude from B to AC is a symmetry:



3. *Translational symmetry* is a symmetry where ϕ is a translation. For example, consider an infinite grid of unit squares in $\mathbb{R}^2$ and a translation $\phi : (x, y) \mapsto (x + 1, y)$, which corresponds to shifting each square to the position of the square to its right. Clearly, ϕ is an symmetry of the grid since $x \mapsto x + 1$ is a bijection in $\mathbb{R}$.
4. *Scale symmetry* is a symmetry where ϕ is a *scale transform*, i.e. $\exists \lambda \in \mathbb{F}$ s.t. $\phi : v \mapsto \lambda v$. For example, consider an infinite line in the Euclidean plane $\updownarrow = (t, 0)$ for $t \in \mathbb{R}$ and a scale transform ϕ, then $\phi(l) = (\lambda t, 0)$. The equation $(t_1, 0) = (\lambda t_2, 0)$ has a solution for every t_1 if $\lambda \neq 0$, so if $\lambda \neq 0$ then ϕ is a scale symmetry of the infinite line. A more exotic example is fractals, which are self-similar and exhibit a form of scale symmetry as we "zoom in".

5. *Glide symmetries* are symmetries where ϕ is a composition of a translation and a rotation. An example is footprints in the sand.

DEFINITION 0.2 $\langle$symmetry of a polygon$\rangle$

Given a polygon P and a mapping ϕ, we say that ϕ is a symmetry of P if ϕ is an isometry *s.t. $\phi(P) = P$.*

This means we exclude transformations which map P to isomorphic shapes, for example if $\phi(\square ABCD) = \square BACD$, meaning that B and C are adjacent in the pre-image but not in the image, then ϕ is not a symmetry of ϕ even though both the image and pre-image overlap. The reason for this is that such transformation does not preserve the *structure* of the polygon.

We now consider the symmetries of a regular n-gon in $\mathbb{R}^2$, although some of the results we prove can be generalized to higher dimensions. First we note that out of the types of symmetries mentioned above, it cannot be that an n-gon has a translational symmetry because a (non-identity) translation would also translate the boundary of the n-gon so the boundaries wouldn't overlap, so the mapping wouldn't be a symmetry, so by Theorem 7 and by Theorem 6 classification of isometries of R^n the two remaining candidates are rotations and translations (and compositions of the two). In the following discussion we investigate the symmetries of an n-gon and conclude with a characterization of those symmetries.

Consider that in both examples of symmetries of an equilateral triangle, the symmetries map adjacent vertices to adjacent vertices. Intuitively we know this must be true in general since we consider the symmetries of P_n as a subset of rigid motions in $\mathbb{R}^2$, and appealing to our analogy from physics we know that rigid bodies do not decompose and recompose under rigid motions so the shape must remain intact throughout the motion. Formalizing this idea without appealing to intuitive ideas of continuous rigid motions is a bit more involved. We do this below:

THEOREM 7

Let P_n be a regular n-gon and ϕ a symmetry of P_n (i.e. $\phi(P_n) = P_n$ and ϕ is an isometry), then if v_1, v_2 are adjacent vertices of P_n then $\phi(v_1), \phi(v_2)$ are also adjacent. More simply put, a symmetry of a regular n-gon preserves the adjacency of the vertices of the n-gon.

Proof. We use the following lemma:

LEMMA 7.1

The boundary of an n-gon P_n is mapped to itself by a symmetry ϕ of P_n, i.e. $\phi(\partial P_n) \subseteq \partial P_n$.

Proof. (lemma) Suppose not, then $\exists y \in \phi(\partial P_n), y \notin \partial P_n$. Now consider the pre-image $x \in \partial P_n$ s.t. $\phi(x) = y$ (there is a unique such x since a symmetry of an n-gon is an isometry and these are injective). Since $x \in \partial P_n$ then by definition $\forall \epsilon 0$ the ϵ-ball centred at x, $B_\epsilon(x)$, contains points both in the interior of P_n and in the exterior of P_n. Also since $\phi(x) \notin \partial P_n$ and $\phi(x) \in P_n$ since $\phi(P_n) = P_n$, then it must be that $B_\epsilon(y) = B_\epsilon(\phi(x)) \subseteq$ Interior(P_n) for a sufficiently small ϵ.

Now take $x_1 \in B_\epsilon(x)$ s.t. $x_1 \notin P_n$, then $\phi(x_1) \in B_e(\phi(x))$ so $\phi(x_1) \in$ Interior(P_n), but also since $\phi(P_n) = P_n$ since ϕ is a symmetry then $\exists x_2 \in P_n$ s.t. $\phi(x_2) = \phi(x_1)$ since $\phi(x_1)$ is in the interior of P_n, so we have $\phi(x_1) = \phi(x_2)$ and $x_1 \neq x_2$, so ϕ is not injective, a contradiction, so $\phi(\partial P_n) \subseteq \partial P_n$.

□

We now prove the main theorem:

(proof cont'd) Let $v_1, v_2 \in \partial P_n$ be adjacent vertices of P_n. By 6 we know that $\phi(v_1), \phi(v_2) \in \partial P_n$. Suppose by contradiction they are not adjacent in the image of P_n, i.e. $\exists v_3 \in \partial P_n$ s.t. $v_3 \notin \updownarrow_{v_2 v_1} = \updownarrow$ (where $\updownarrow_{v_2 v_1} = \updownarrow$ is the edge joining v_1 and v_2) and $\phi(v_3) \in \updownarrow_{\phi(v_2)\phi(v_1)} = \updownarrow_\phi$.

Denote $y = \phi(v_3)$, then y is the point on $\updownarrow_\phi$ in distance a from v_2 and b from v_1. Consider the point on $\updownarrow$ in distance a from v_2 and b from v_1, and denote it x. Since ϕ is an isometry then it must be that $d(\phi(x), \phi(v_2)) = d(x, v_2)$ and likewise for x, v_1 and their images, but then $\phi(x)$ is at the same distance from $\phi(v_1), \phi(v_2)$ as y, and and since $y \in \updownarrow_\phi$ this is only possible if $y = \phi(x)$, so it follows that x, v_3 are both pre-images of y, but ϕ is an isometry so it is injective, so $x = v_3$, but then $v_3 \in \updownarrow_{v_1 v_2}$ so it cannot be a vertex of P_n, a contradiction, so ϕ maps adjacent vertices of P_n to adjacent vertices.

□

Equipped with this result, it follows immediately that if ϕ is a symmetry of P_n then it is uniquely determined by its action on a pair of adjacent vertices of P_n. Informally this is true since by fixing a pair of vertices, we also determine where the other adjacent vertices of those vertices will be mapped to, since those must maintain adjacency with the fixed vertices, and this process is then repeated until the position of all vertices have been determined (of course, all of them will be on the boundary of P_n as per 6). We formalize this observation below:

PROPOSITION 6

Let P_n be an n-gon and $3 \leq n$, and $\{v_1, v_2\} \in E(P_n)$ (i.e. they are adjacent vertices). If ϕ is a symmetry of P_n, then it is uniquely determined by its action of v_1, v_2, i.e. $\phi(v_1), \phi(v_2)$ uniquely define ϕ.

Proof. Suppose not, then there exist two different symmetries $\phi, \psi, \phi \neq \psi$ of P_n s.t. $\phi(v_1) = \psi(v_1) = w_1$ and $\phi(v_2) = \psi(v_2) = w_2$. By 7.1 we know that since $\{v_1, v_2\} \in E(P_n)$ then $\{\phi(v_1), \phi(v_2)\} \in$

$E(\phi(P_n))$ and $\{\psi(v_1), \psi(v_2)\} \in E(\psi(P_n))$.

If ϕ, ψ agree $\forall v \in V(P)$, then it must be that $\phi = \psi$, since both are isometries so they map lines to lines, so they map edges to edges, and the vertices are the intersections of those edges, so if both mappings agree on the vertices, then they must agree on the intersection of edges, and since edges are mapped to other edges (by 6) it must be that ϕ, ψ map edges to the same edges, so $\phi = \psi$. Since by supposition $\phi \neq \psi$, then $\exists w \in V(P)$ s.t. $\phi(w) \neq \psi(w)$. We will prove by induction on n that there cannot exist such w, a contradiction, so $\phi = \psi$.

First, for $n = 3$ we have that w must adjacent to both v_1 and v_2 (since P_3 is simply the image of the linear embedding of the geometric realization of the cycle graph C_3). By 6 ϕ, ψ map w to a vertex of P_3, and since both are isometries they are injective so it cannot be that w is mapped to the same vertex as v_1 or v_2, which leaves only one vertex w can be mapped to, and since ϕ, ψ agree on the mapping of v_1, v_2, then it must be that they agree on the mapping of w since both have the same option left, so for $n = 3$ we have $\phi = \psi$.

Suppose by induction the theorem holds for some n, and consider $n + 1$. By the two ears theorem we know that P_{n+1} has at least two vertices which can be removed without introducing crossings. Removing one such vertex breaks P_{n+1} down into two polygons, one with n vertices and one with 3 vertices, so the induction hypothesis holds for both polygons and ϕ, ψ must agree on all the vertices of those polygons, so it must be that they agree on all vertices of P_{n+1}, which completes the proof by induction.

We conclude via induction that if ϕ, ψ agree on a pair of adjacent vertices, then $\forall v \in V(P), \phi(v) = \psi(v)$, so $\phi = \psi$, so the symmetry is uniquely determined via its action of two adjacent vertices.

$\square$

Consider the number of symmetries of a regular n-gon. In our example of an equilateral triangle $\triangle ABC$, we quickly identify 6 symmetries:
1. Identity
2. Rotation around the center by 120 CCW
3. Rotation around the center by 240 CCW
4. Reflection around the altitude from A to CB
5. Reflection around the altitude from B to AC
6. Rotation around the altitude from C to AB

We see that for $n = 3$, we identify 6 different symmetries (to see they are different, simply demonstrate that no two symmetries map A, B, C to the same vertices). As you may have guessed, it is true in general that every n-gon has at most $2n$ distinct symmetries, as we prove below:

THEOREM 8

An n-gon P_n has at most $2n$ distinct symmetries.

Proof. Suppose ϕ is a symmetry of P_n, then by Theorem 8 it is uniquely determined by where it sends a pair of adjacent vertices $\{v_1, v_2\} \in E(P_n)$. As we have argued, in the proof for that proposition, it sends vertices to vertices. Consider v_1 under ϕ. Since P_n has n vertices and ϕ sends v_1 to a vertex, there are n possible choices for $\phi(v_1)$. By 7.1 , the adjacent vertices are mapped to adjacent vertices, so if $3 \leq n$ then the vertex $\phi(v_1)$ has exactly 2 distinct adjcent vertices, so there are exactly 2 choices for $\phi(v_2)$, so in total there are $n \times 2$ possible mappings that could be symmetries of P_n, so P_n has at most $2n$ distinct symmetries.

□

If P_n is a regular n-gon, then all $2n$ candidates are in fact symmetries of P_n since all sides are equal, so it does not matter to which edge v_1v_2 is sent so long as the mapping maintains the boundary and sends all vertices of P_n to vertices of P_n while maintaining adjacency. We write this as a corollary to the theorem:

COROLLARY 8.1

If P_n is a regular n-gon, then P_n has exactly $2n$ distinct symmetries.

We are now primed to classify all symmetries of a regular n-gon.

THEOREM 9 $\langle$classification of symmetries of a regular n-gon$\rangle$

Let P_n be a regular n-gon, then:
1. P_n has n distinct purely rotational symmetries $\{r, r^2, \ldots, r^n\}$.
2. Given a reflection symmetry f of P_n, $f^2 = e$ where e is the identity symmetry.
3. All symmetries of P_n be can expressed in terms of a single reflection symmetry f and its n rotational symmetries $\{r, r^2, \ldots, r^n\}$.

Proof. 1. Rotating P_n around its center n times by $\frac{2\pi}{n}$ results in a total rotation of 2π about its center, which returns P_n to its original position, so if e is the identity, r is a rotation by $\frac{2\pi}{n}$ and r^k is repeating r k times, then $r^n = e$. This means that P_n has n distinct *rotational symmetries*.

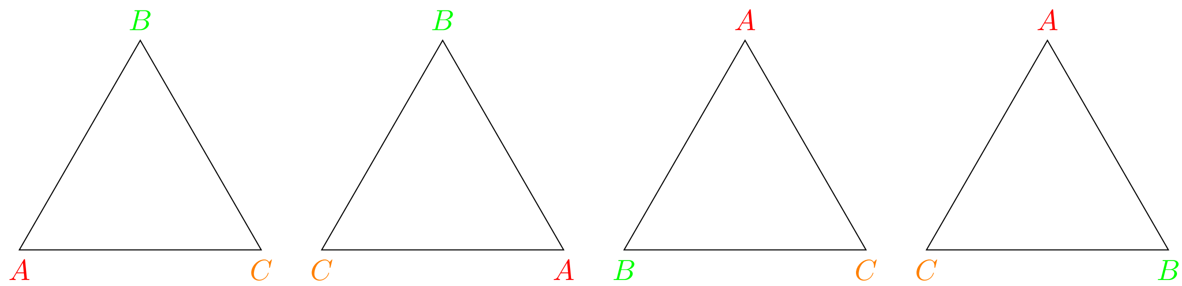
2. If f is a reflection symmetry of P_n, then reflecting P_n twice around the same line restores P_n to its starting position, i.e. $f^2 = e$.

3. Since reflections inverts the orientation of P_n, and since P_n has n distinct rotational symmetries and $2n$ distinct symmetries in general, and all rotations preserve orientations, then given a (non-identity) rotation r^k and a reflection f, fr^k is a new symmetry (since it has inverted orientation compared to all purely rotational symmetries of P_n). Composing f with each rotation $\{r, r^2, \ldots, r^n\}$, we construct a new set of n distinct symmetries $\{fr, fr^2, \ldots, fr^n\}$ (arguing that each fr^k is unique is left as an exercise). In conclusion, we have a set of $2n$ distinct symmetries, and by Theorem 9 classification of symmetries of a regular n-gon this is exactly the set of symmetries of P_n.

□

By the theorem, it suffices to consider a single reflection symmetry f of P_n when constructing all symmetries of P_n. We now turn our attention to the final relation which we will use to define the symmetry group of a regular n-gon.

Consider $\triangle ABC$ whose center is O. As we have seen, rotating it by $\frac{2\pi}{3}$ degrees around O CCW produces a symmetry. Trivially, rotating it by $\frac{2\pi}{3}$ degrees around O CW also produces a symmetry (identify that this is the same as rotating the triangle by $2\frac{2\pi}{3}$ degrees around O CCW). Now, suppose $\triangle ABC$ is reflected about the line BO, which as we have seen before produces a symmetry which inverts the orientation of the triangle - it maps A to C and C to A, while fixing B. Suppose we now rotate the triangle by $\frac{2\pi}{3}$ degrees CCW, which will again result in a symmetry, and finally we reflect again around the line formed by BO in the original triangle (i.e., applying the same reflection, disregarding where BO is mapped to under the sequence of symmetries). We denote the reflection f and the rotation r, then the symmetry we constructed is the composition of f, r and finally f again, i.e. it is frf (where composition is juxtaposed, note that since f and r are both origin-preserving isometries they both correspond to matrices, so frf can also be interpreted as *multiplication* of the corresponding transformation matrices). This process is illustrated below:



left to right: initial triangle, then reflected around BO, then rotated by $\frac{2\pi}{3}$ CCW, then reflected again

Identify that this is *exactly* the same as r^{-1}, i.e. rotating the initial triangle by $\frac{2\pi}{3}$ degrees clockwise, thus we observe that $frf = r^{-1}$ for an equilateral triangle. It is easy to extend this argument to any regular n-gon:

THEOREM 10

Let P_n be a regular n-gon and r be a rotational symmetry of P_n and f be a reflection symmetry of P_n, then $frf = r^{-1}$ (where r^{-1} is a rotation of P_n by $\frac{2\pi}{n}$ degrees in the opposite direction of r. here we used CCW as the direction of rotation, so r^{-1} rotates CW).

Proof. If n is odd and f is a reflection symmetry of P_n then f must pass through one of the vertices of P_n, otherwise if n is even then either f passes through a vertex of P_n or passes through the midpoint of two parallel edges of P_n. By Theorem 10 by any choice of f we can construct the symmetries of P_n, so it suffices to demonstrate the theorem holds for an arbitrary reflection symmetry f of each of the aforementioned types.

Suppose n is odd, then f is a reflection around a line $\updownarrow$ which passes through a vertex $v \in V(P_n)$. Consider the adjacent vertices $u, w \in V(P)$ of v, w.l.o.g suppose that the order of vertices (going CCW) is $u \to v \to w$. Since f inverts orientation, and f passes through v, then f maps u, v, w as follows: $f : u \mapsto w, v \mapsto v, w \mapsto u$. We now compose f with a rotation r, which moves each vertex to the next vertex in a CCW fashion, i.e. $r : u \mapsto u', v \mapsto u, w \mapsto v$ (where u' the adjacent vertex to the other side of u. If $n = 3$, then $u' = w$), so $rf(u) = r(f(u)) = r(w) = v, rf(v) = r(v) = u, rf(w) = r(u) = u'$. Finally, we compose with f again and get $frf(u) = f(rf(u)) = f(v) = v, frf(v) = f(rf(v)) = f(u) = w$, so frf maps the edge uv to the edge vw which uniquely determines the symmetry frf by Theorem 8 , but this is exactly rotation by $\frac{2\pi}{3}$ clockwise (u, v were both moved to the vertex clockwise next to them), so $frf = r^{-1}$.

If n is even, then we also need to consider the case where the line of symmetry of f passes through an edge and not a vertex. In this case, consider such edge (there are at least 2 such edges since the line passes through the n-gon) and let u, v be the vertices at the ends of the edge where v follows u in a CCW order, and w be the next vertex after v in a CCW order (i.e. the order is again $u \to v \to w$, with the line of symmetry passing through the edge uv). Under f, $u \mapsto v, v \mapsto u, w \mapsto u'$ where u' the the vertex preceding u CCW (i.e. $u' \to u \to v \to w$) , and under r we have $u' \to u, u \to v, v \to w$. Consider the adjcent vertices v, w under frf, we have $frf(v) = fr(u) = f(v) = u$, and $frf(w) = fr(u') = f(u) = v$, so we have that frf maps v to u and w to v, i.e. it shifts v, w to the next vertex in a clockwise order which is exactly the same as applying r^{-1} on both vertices, and by Theorem 8 this uniquely determines the symmetry, so $frf = r^{-1}$.

□

Dihedral Group

Now that we have provided enough geometrical background, we can finally present the algebraic structure of the group of symmetries of a regular n-gon P_n, called the *dihedral group* of order $2n$, denoted D_{2n}.

DEFINITION 0.3 $\langle$dihedral group$\rangle$

The dihedral group of order $2n$, D_{2n}, is given by $D_{2n} = \langle r, f | f^2 = e, r^n = e, frf = r^{-1}\rangle$ (i.e. it is tihe group generated by r, f under multiplication, given the set of relations $f^2 = e, r^n = e, frf = r^{-1}$ meaning 2 is the order of f, n is the order of r, and $frf = r^{-1}$).

Note that this definition is independent of the geometric interpretation of the elements of D_{2n} as symmetries of a regular n-gon, but by Theorem 10 and 0.2 dihedral group we see that the symmetries of a regular n-gon can be thought of as the elements of D_{2n}.

EXAMPLE 10.1

For $n = 3$, D_6 is the dihedral group whose elements are $\{e, r, r^2, f, fr, fr^2\}$. This corresponds with our geometrical results that a regular n-gon has $2n$ symmetries which can be written as n distinct rotations $\{r, r^2, \dots, r^n = e\}$ and the composition of a reflection with each rotation.

We usually identify D_{2n} with the set of symmetries of a regular n-gon, so as per Theorem 10 $D_{2n} = \{e, r, \dots, r^{n-1}, f, fr, \dots, fr^{n-1}\}$. We will demonstrate that this set is indeed a group under composition:

LEMMA 10.1

In a dihedral group, $fr^k f = r^{-k}$.

Proof. Via our geometrical interpretation of D_{2n} we know this is true since a reflection followed by a rotation followed by the same reflection geometrically inverts the direction of that rotation as we have argued above, however to properly prove this property for D_{2n} we need to refer to the group definition.

We prove by induction of k. If $k = 0$ then $fr^0 f = fef = ff = f^2 = e = r^{-0}$ (since the order of f in D_{2n} is 2) and if $k = 1$ then $fr^1 f = frf = r^{-1}$ (by the relation $frf = r^{-1}$). Suppose the lemma holds for some k, and consider $k + 1$, then by definition $r^{k+1} = r^k r$, and by the properties of the identity and associativity $r^k r = r^k er$, finally since $f^2 = e$ we substitute e for $f^2 = ff$ and have $r^{k+1} = r^k ffr$. Pre and post multiplying both sides by f, we have $fr^{k+1} f = fr^k ffrf = (fr^k f)(frf)$. By the induction hypothesis $fr^k f = r^{-k}$ and by definition of D_{2n}, $frf = r^{-1}$, so $fr^{k+1} f = r^{-k} r^{-1} = r^{-k-1} = r^{-(k+1)}$, which concludes the proof by induction.

□

PROPOSITION 7

The dihedral group $D_{2n} = \{e, r, \ldots, r^{n-1}, f, fr, \ldots, fr^{n-1}\}$ is a group under composition.

Proof. 1. (identity) $e \in D_{2n}$ is the identity under composition.
3. (associativity) function composition is associative.

To demonstrate that D_{2n} is closed under composition and that $\forall a \in D_{2n}(a^{-1} \in D_{2n})$ (inverse), we consider all possible cases:

Closure: Let $a, b \in D_{2n}$.
1. If one of a, b is e, w.l.o.g $a = e$, then $ab = eb = b \in D_{2n}$.
2. If $a = r^k$ and $b = r^m$, $m, k \in \mathbb{N}, m, k \in [0, n-1]$, then $ab = r^k r^m = r^{k+m}$, and by the Euclidean division algorithm we know that $\exists! t, q \in \mathbb{Z}, q \in [0, n)$ s.t. $k + m = tn + q$, so $r^{k+m} = r^{tn+q} = r^{tn}r^q = (r^n)^t r^q$, and since $r^n = e$ then $(r^n)^t r^q = r^q$ with $q \in [0, n)$, so $r^q \in D_{2n}$.
3. If $a = r^k$ and $b = fr^m$, $m, k \in \mathbb{N}, m, k \in [0, n-1]$ then $ab = r^k fr^m$. Since $e = f^2 = ff$ we can write $ab = eab = ffab = ffr^k fr^m = f(fr^k f)r^m$. Now by 7 we have $f(fr^k f)r^m = fr^{-k}r^m = fr^{m-k}$. Again we write $m - k = tn + q$ with $q \in [0, n)$, so $ab = fr^{m-k} = fr^{tn+q} = fr^q$, and since $q \in [0, n)$ then $fr^q \in D_{2n}$.
4. If $a = fr^m$ and $b = r^k$, $m, k \in \mathbb{N}, m, k \in [0, n-1]$ then $ab = fr^m r^k = fr^{m+k}$, and since we have already demonstrated that $r^{m+k} = r^q$ with $q \in [0, n)$ then $fr^{m+k} = fr^q \in D_{2n}$.

We conclude that D_{2n} is closed under composition.

Inverse: Let $a \in D_{2n}$.
1. If $a = e$, then $a^{-1} = a = e \in D_{2n}$.
2. If $a = r^k$, $k \in \mathbb{N}, k \in [0, n-1]$, then by 7 $r^{-k} = fr^k f$ which is in D_{2n} due to closure of D_{2n} under compostion.
3. If $a = f$, then $aa = ff = f^2 = e$, so $a^{-1} = a \in D_{2n}$.
4. If $a = fr^k$, $k \in \mathbb{N}, k \in [0, n-1]$ then $(r^{-k}f)(fr^k) = r^{-k}(ff)r^k = r^{-k}f^2 r^k = r^{-k}er^k = r^{-k}r^k = r^{-k+k} = r^0 = e$, so $r^{-k}f$ is the inverse of fr^k, and since we demonstrated that $r^{-k} \in D_{2n}$ and that D_{2n} and we know that $f \in D_{2n}$ and that D_{2n} is closed under multiplication, then $r^{-k}f \in D_{2n}$.

We conclude that $\forall a \in D_{2n}(\exists a^{-1} \in D_{2n}(aa^{-1} = a^{-1}a = e))$.

We have that the structure $(D_{2n}, \circ)$ is a set D_{2n} with a binary operator $\circ$ on the set that is associative and has the identity and inverse properties, so $(D_{2n}, \circ)$ is a group.

□

Considering the symmetries of a regular n-gon in algebraic terms makes computations involving these operations much simpler. For example, determining where each vertex of P_6 will go under $fr^5fr^{-1}f^2$ appears non-trivial at first, but using the relations of D_{2n} and the fact that it is a group, we are able to immediately say that the sequence $fr^5fr^{-1}f^2$ results in a symmetry of P_6 (closure under the group operation) and that it is the same as either r^k for some non-negative integer k less than 6, or fr^k, and we can even calculate which of the $2n = 2 \cdot 6 = 12$ symmetries of P_6 it corresponds to:

$$fr^5fr^{-1}f^2 = (fr^5f)r^{-1}e = r^{-5}r^{-1} = r^{-6} = r^{-1\cdot 6} = (r^6)^{-1} = e^{-1} = e$$

So we conclude that applying the sequence of symmetries $fr^5fr^{-1}f^2$ to a regular hexagon has no effect on the hexagon. Demonstrating this geometrically would've been a huge undertaking.

We will revisit dihedral groups in discussions about group theory and abstract algebra.